# RIO: Runtime Intelligence Orchestration

## A Technical White Paper on Governed AI Execution

**Author:** Brian K. Rasmussen
**Date:** March 2026
**Repository:** github.com/bkr1297-RIO/rio-system

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## 1. Abstract

Runtime Intelligence Orchestration (RIO) is a fail-closed authorization and audit runtime that governs the execution of actions performed by autonomous AI agents. RIO decouples the intelligence of an AI system from the authority to execute real-world actions by inserting a structural governance layer between the agent and the tools it controls. Every action follows a deterministic pipeline: intent is created, policy is evaluated, human approval is obtained when required, execution is gated, outcomes are verified, and a cryptographically signed receipt is appended to a tamper-evident ledger. The system enforces governance structurally at runtime rather than relying on model alignment, prompt engineering, or advisory guardrails.

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## 2. Problem Statement

Large Language Models have evolved from conversational interfaces into autonomous agents capable of sending emails, transferring funds, modifying infrastructure, and accessing sensitive data through API connectors and tool-use frameworks. This transition introduces a fundamental asymmetry: AI can propose and attempt actions at machine speed, while human oversight operates at human speed.

Current approaches to controlling AI agent behavior fall into three categories, all of which are insufficient for production environments:

| Approach | Mechanism | Failure Mode |
|---|---|---|
| **Prompt Engineering** | System instructions telling the AI to “be careful” | Advisory only; overridden by prompt injection or model drift |
| **Model Alignment** | Training-time reinforcement of desired behaviors | Probabilistic; no runtime guarantee of compliance |
| **UI Confirmation** | Frontend dialog asking user to click “Confirm” | Client-side only; removable by any party with platform access |

None of these approaches provide structural enforcement. They rely on the AI’s willingness to follow rules, the stability of trained behaviors, or the integrity of a client-side interface. In a production environment where an AI agent holds OAuth tokens to Gmail, GitHub, or a payment processor, the absence of a server-side execution gate means that a single prompt injection, a platform configuration change, or a model update can result in unauthorized actions with no audit trail.

The core problem is architectural: the tools that give AI agents the ability to act in the real world have no built-in governance layer. The connection between an AI agent and an external API is a direct pipe with no intermediary enforcing authorization, logging, or verification.

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## 3. Solution: Governed AI Execution

RIO addresses this gap by introducing a governed execution runtime that sits between AI agents and the external systems they interact with. RIO is not an AI model, not a chatbot framework, and not a monitoring dashboard. It is a control plane through which every AI-initiated action must pass before reaching the target system.

The runtime enforces three guarantees:

1. **No execution without authorization.** The system defaults to blocking all actions. Execution proceeds only after the governance pipeline produces a valid, signed approval. This is a fail-closed architecture: if any component is unavailable, missing, or invalid, execution is denied.
2. **Every action produces a cryptographic receipt.** Each completed action generates a receipt containing hashes of the original intent, the executed action,

and the verified outcome. Receipts are signed and appended to a hash-chained ledger, creating a tamper-evident audit trail.

3. **Governance is structural, not advisory.** Authorization is enforced server-side at the execution gateway. The AI agent cannot bypass the gate by modifying its own instructions, exploiting a prompt injection, or accessing tools directly. All external API keys are stored in the gateway environment variables; the agent does not possess them.

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## 4. Core Loop: Intent → Governance → Execution → Receipt

RIO operates on a single deterministic pipeline that every action must traverse. The pipeline consists of four stages:

### 4.1 Intent

The AI agent or an upstream system submits a structured intent describing the proposed action. The intent includes the action type (e.g., `send_email`, `create_file`), the target system, the parameters, and the requesting identity. Once submitted, the intent is immutable and assigned a unique identifier.

### 4.2 Governance

The governance stage evaluates the intent against the active policy set. The policy engine classifies the action by risk level and returns one of three verdicts:

- **AUTO_APPROVE** — The action falls within pre-authorized boundaries. Execution proceeds without human intervention.
- **REQUIRE_HUMAN** — The action exceeds autonomous thresholds. Execution is blocked until a human authority reviews and approves the intent.
- **AUTO_DENY** — The action violates a hard policy constraint. Execution is permanently blocked.

For actions requiring human approval, the system presents the intent to the designated authority through the Gate UI. The authority reviews the proposed action

and issues an approval or denial. Approvals are cryptographically signed (using Ed25519), time-limited, and bound to the specific intent identifier.

### 4.3 Execution

The execution gateway receives the approved intent and performs verification checks before allowing the action to proceed:

1. A valid approval record exists and its Ed25519 cryptographic signature is verified.
2. The approval has not expired.
3. The approval has not been previously consumed (single-use enforcement).
4. The execution parameters match the original intent exactly.

If all checks pass, the gateway dispatches the action to the appropriate connector and records the execution result. If any check fails, execution is denied.

### 4.4 Receipt

Immediately following execution, the system generates a cryptographic receipt that binds together the intent, the approval, and the execution outcome. The receipt is hashed, signed, and appended to the ledger.

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## 5. System Architecture

RIO consists of core components that enforce the separation of intelligence from execution.

| Component | Responsibility |
|---|---|
| **Intent Service** | Receives and records proposed actions as immutable intent objects |
| **Policy Engine** | Evaluates intents against the active policy set |
| **Gate UI (ONE)** | Presents pending intents to human authorities for review and approval |
| **Execution Gateway** | Verifies approvals, enforces single-use tokens, dispatches actions to connectors. Holds all external API keys. |
| **Connectors** | Execute actions against external systems (Gmail, Drive, GitHub) |
| **Ledger** | Stores signed receipts in a hash-chained, append-only audit log (PostgreSQL) |

The architecture dictates that **all actions must pass through the RIO Gateway. No agent may call external APIs directly.**

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# 6. Receipt Protocol

The receipt is the atomic unit of accountability in RIO. It is the cryptographic proof that an action was requested, authorized, executed, and recorded.

## 6.1 Hash Computation

Three hashes are computed for each governed action:

- `intent_hash`: SHA-256 of the canonicalized intent JSON. Proves the original request has not been altered.
- `action_hash`: SHA-256 of the canonicalized execution parameters. Proves the executed action matches what was authorized.
- `verification_hash`: SHA-256 of the canonicalized execution result. Proves the outcome was recorded accurately.

### 6.2 Chain Integrity

Each receipt's `previous_hash` field creates a backward link to the preceding receipt. This creates a structure analogous to a blockchain: any modification to a historical receipt invalidates all subsequent hashes, making tampering mathematically detectable.

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## 7. Security Model (Fail Closed, Human Approval)

RIO's security model is built on the principle that AI agents are untrusted requesters.

### 7.1 Fail-Closed Architecture

The default state of the execution gateway is **deny**. If any condition fails (missing approval, invalid signature, expired token, parameter mismatch), execution is blocked. The system does not degrade gracefully into permissive mode.

### 7.2 Human Approval and Identity Binding

No high-risk execution occurs without explicit human approval. Approvals are bound to an authenticated user identity (via OAuth) and must include an Ed25519 signature. The receipt permanently records the identity of the human who authorized the action.

### 7.3 Credential Isolation

The AI agent never holds OAuth tokens or API keys directly. All credentials (OpenAI, Anthropic, Google, GitHub) are stored securely in the gateway's environment variables. The agent can only submit intents to the governance pipeline.

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## 8. Current Implementation

The RIO system currently consists of the following working components:

- **Execution Gateway:** Express.js and Python FastAPI implementations that enforce the governance pipeline.

- **Connectors:** TypeScript adapters for Gmail, Google Drive, and GitHub.
- **Receipt Protocol:** Formal JSON schemas for ActionRequest, AIRecommendation, ApprovalRecord, ExecutionRecord, and Receipt.
- **Verification Suite:** Automated security tests proving the fail-closed nature of the gateway ($^{10}/_{10}$ passing).
- **Demo UI:** Interactive frontend demonstrating the approval flow.

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## 9. Deployment Architecture (Azure)

To transition from a prototype to a permanent, secure, $^{24}/_{7}$ runtime environment, RIO targets the following deployment architecture on Microsoft Azure:

1. **Azure App Service:** Hosts the FastAPI/Express backend gateway and serves the frontend static files. Ensures $^{24}/_{7}$ uptime.
2. **Azure Database for PostgreSQL:** Serves as the persistent, append-only ledger for storing hash-chained receipts.
3. **Azure Key Vault:** Securely stores the Ed25519 private keys, HMAC signing keys, and all external API tokens (Gmail, Drive, GitHub).
4. **Azure Entra ID (OAuth):** Provides identity binding, ensuring that only the authenticated root authority (Brian Kent Rasmussen) can sign and approve intents.
5. **GitHub Actions:** Provides Continuous Deployment (CI/CD) from the `bkr1297-RIO/rio-system` repository to the Azure App Service.

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## 10. Use Cases

RIO applies to any scenario where an AI agent performs actions with real-world consequences:

- **Governed Email:** An AI drafts an email. RIO intercepts the send action, requires human approval, and embeds the cryptographic receipt in the email body.

- **Financial Operations:** A finance agent proposes a payment. RIO enforces thresholds (e.g., payments over $1,000 require manager approval) and records the exact amount and approver in the ledger.
- **Infrastructure Management:** A DevOps agent proposes a production deployment. RIO requires director-level approval and logs the commit hash and deployment target.
- **Subscription Management:** An AI identifies unused subscriptions and proposes cancellation. RIO presents the cancellation for approval and files a receipt documenting the action.

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## 11. Definitions

| Term | Definition |
|---|---|
| **RIO** | Runtime Intelligence Orchestration. A fail-closed governance runtime for AI agent execution. |
| **Intent** | A structured, immutable record of a proposed action submitted by an AI agent. |
| **Execution Gateway** | The server-side enforcement point that verifies approvals and dispatches actions to connectors. |
| **Connector** | An execution module that interfaces with an external system (Gmail, Drive, GitHub, etc.). |
| **Receipt** | A cryptographically signed record binding together the intent, approval, execution, and outcome. |
| **Ledger** | An append-only, hash-chained audit log of all receipts. |
| **Fail-Closed** | A design principle where the default state is deny; execution requires explicit authorization. |

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## 12. Conclusion

RIO introduces governed execution as a new architectural layer for AI systems. It is not an alternative to model alignment or prompt engineering — it is a complementary structural layer that enforces governance at runtime regardless of the AI model's internal state.

The core insight is that AI agents should be treated as untrusted requesters. The solution is to validate everything server-side, enforce authorization structurally, and maintain an immutable audit trail. Governed execution does not slow AI systems down; it makes them accountable. Every action is authorized, executed, verified, recorded, and available for audit. The system enforces the rules — not the AI.